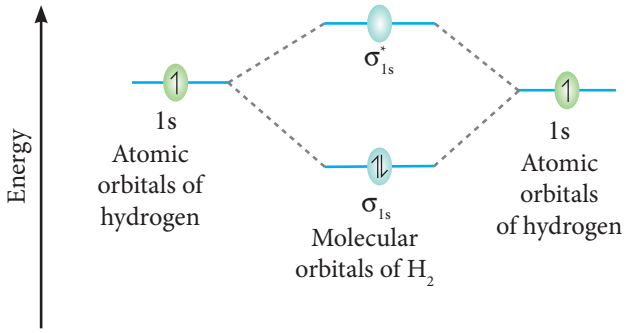
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10.10.2 Bonding in some Homonuclear di-atomic molecules:



Molecular orbital diagram of hydrogen molecule (H_2)

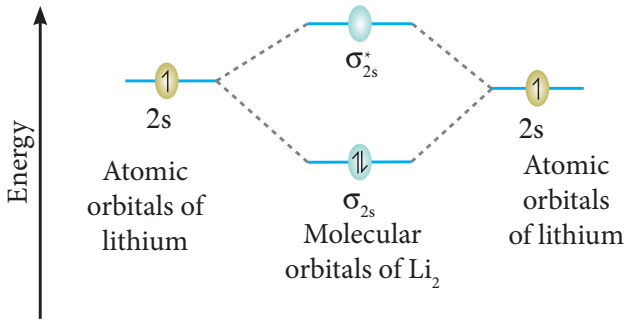
Electronic configuration of H atom $1s^1$

Electronic configuration of H_2 molecule σ_{1s}^2

Bond order = $\frac{N_b - N_a}{2} = \frac{2-0}{2} = 1$

Molecule has no unpaired electrons hence it is diamagnetic.

Fig 10.30 MO Diagram for H_2 molecule



Molecular orbital diagram of lithium molecule (Li_2)

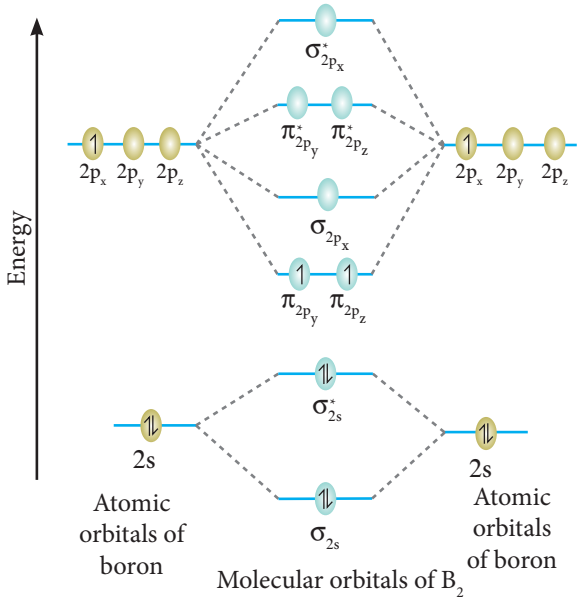
Electronic configuration of Li atom $1s^2$ $2s^1$

Electronic configuration of Li_2 molecule $\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2$

Bond order = $\frac{N_b - N_a}{2} = \frac{4-2}{2} = 1$

Molecule has no unpaired electrons hence it is diamagnetic.

Fig 10.31 MO Diagram for Li_2 molecule



Molecular orbital diagram of boron molecule (B_2)

Electronic configuration of B atom $1s^2$ $2s^2$ $2p^1$

Electronic configuration of B_2 molecule $\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2, \sigma_{2s}^{*2}, \pi^1_{2p_y}, \pi^1_{2p_z}$

Bond order = $\frac{N_b - N_a}{2} = \frac{6-4}{2} = 1$

Molecule has two unpaired electrons hence it is paramagnetic.

Fig 10.32 MO Diagram for B_2 molecule